

9(a)	work done per unit charge	B1
	(work done) moving positive charge from infinity	B1
9(b)(i)	energy = $4.8 \times 1.60 \times 10^{-13}$ $= 7.7 \times 10^{-13} \text{ J}$	A1
9(b)(ii)	$E_P = Qq / 4\pi\epsilon_0 d$	C1
	$Q = 79e$ and $q = 2e$	C1
	$7.68 \times 10^{-13} = (79 \times 2 \times \{1.60 \times 10^{-19}\}^2 / (4\pi \times 8.85 \times 10^{-12} \times d)$	C1
	$d = 4.7 \times 10^{-14} \text{ m}$	A1
9(c)	(diameter must be) less than/equal to 10^{-13} or 10^{-14} m	B1